Biomedical Knowledge Graph — Technical Documentation

22 node labels · 32 relationship types · 77 source files · 13.7M nodes / 16.8M relationships

1. Architecture

The graph is built **to files, not to a database**. Every loader writes CSV; a validator reads those CSVs; only a build that passes validation is imported. That ordering is the design's core: a bulk import replaces the store outright with no transaction, so an unchecked build silently becomes the live graph and the failure mode is a confident wrong answer rather than an error.

```
S3 (moine-data)
   |  77 declared CSVs
   v
build.py            stream each file, resolve names, emit nodes/edges
   v
~/graph-runs/<ts>/  nodes/*.csv  edges/*.csv  manifest.json
   v
validate.py         referential integrity, key collisions, fixtures
   v  (exit 0 required)
stage_for_neo4j.py  strip headers, collapse newlines, enforce types
   v
neo4j-admin import  -> Neo4j 'biolyt'
   v
schema.cypher       constraints, indexes, 3 full-text indexes
```

Nothing in the build talks to Neo4j. The same CSVs a human inspects are the ones imported, so every claim about the graph is a claim about a table that can be checked without infrastructure.

2. Identity and keys

Every node key is `NAMESPACE:VALUE` and globally unique. `neo4j-admin` resolves all endpoints in one id space, so two labels sharing a key become one node. That happened: `MESH:D000544` was both a Disease and an Identifier until every identifier was routed through a single `Writer.identifier()` that prefixes `ID:`.

Label	Example key	Source of the value
Substance	`UNII:36TN91XZ0V`	gsrs UNII; falls back to CHEMBL: then NAME:
Product	`CA:02245678`	the agency's own licence number
ClinicalTrial	`NCT:NCT01045135`	registry id, canonicalised by trial_key()
Disease	`MESH:D002289`	MeSH descriptor; ICD11: where no MeSH match
Target	`UNIPROT:P04035`	UniProt accession
Variant	`CLINVAR:12345`	ClinVar VariationID, or COSMIC MutationID
Identifier	`ID:UNII:36TN91XZ0V`	always ID:-prefixed, so it cannot collide with the entity it identifies

> **Why trial keys repeat themselves.** 933,232 of 1,048,841 look like `NCT:NCT01045135`, because 19 of 22 registries already embed their prefix in the id. One rule applied uniformly beats a conditional one that breaks when registry 23 arrives, and EUCTR/CTIS need the namespace regardless: `2004-000010-11` identifies nothing on its own.

3. Name resolution

Product ingredient lists, trial interventions and event subjects are free text. A tiered resolver maps them to substance keys, and **every edge records which tier matched** in `match_method`, so a stated fact and an inferred one are never confused downstream.

Tier	Rule	Precedence
`unii`	exact match on the folded name	highest
`salt`	salt/hydrate tokens stripped — *atorvastatin calcium* → atorvastatin	
`stereo`	stereo prefix stripped; built only after every name is known	
`synonym`	ChEMBL alias tier	
`provisional`	no match — a `NAME:` key, visibly unresolved	lowest

> **A false merge is silent and permanent; a missed merge is two visible nodes.** The stereo tier is built only after all names load, and any key two different substances both reduce to is blocked — otherwise *Levo-cetirizine* merges into cetirizine, which are different drugs. 35,692 keys are blocked this way.
>
> Free prose never creates nodes. A product ingredient list may mint a provisional substance because it is a short controlled field; 1.6M trial interventions may not.

4. Node labels

Label	Properties (plus source, run_id on every row)
AdverseEvent	key, term
Approval	key, date, type, status, agency
ClinicalTrial	key, registry, title, status, phase, study_type, enrollment, start_date
Company	key, name, raw_names
Country	key, iso2, name
Disease	key, name, synonyms, vocabulary, tree_numbers
DrugClass	key, atc_code, name, level, vocabulary
Exclusivity	key, code, date, definition
Identifier	key, scheme, value
Mechanism	key, name, action_type
Modality	key, name
OrganClass	key, name
Patent	key, patent_no, expire_date, use_code, use_definition, drug_substance_flag, drug_product_flag
Product	key, name, brand_name, agency, status, form, strength
Publication	key, title, year, journal, doi, pmid, source_db, preprint
Region	key, name
RegulatoryAgency	key, code, name, country, region
RegulatoryEvent	key, type, name, status, reason, start_date, end_date, url
Route	key, name
Substance	key, name, norm_name, synonyms, substance_class, status, max_phase, resolved_by
Target	key, symbol, name, organism, target_type
Variant	key, name, variant_type, gene_symbol, clinical_significance, catalogue, consequence

5. Relationship types

Type	Properties beyond the endpoints
`ABOUT`	`match_method`
`APPROVED_BY`	`match_method`
`APPROVED_IN`	`match_method`
`ASSOCIATED_WITH`	`match_method` , `score`
`BIOSIMILAR_OF`	`match_method`
`CONDUCTED_IN`	`match_method`
`CONTAINS`	`match_method`
`DEVELOPS`	`match_method`
`HAS_ADVERSE_EVENT`	`match_method` , `report_count` , `serious_count` , `death_count`
`HAS_APPROVAL`	`match_method`
`HAS_EXCLUSIVITY`	`match_method`
`HAS_IDENTIFIER`	`match_method`
`HAS_MECHANISM`	`match_method`
`HAS_MODALITY`	`match_method`
`HAS_ROUTE`	`match_method`
`IMPLICATED_IN`	`match_method` , `significance`
`INDICATED_FOR`	`match_method`
`IN_CLASS`	`match_method`
`IN_ORGAN_CLASS`	`match_method`
`IN_REGION`	`match_method`
`ISSUED_BY`	`match_method`
`IS_SALT_OF`	`match_method`
`MENTIONS`	`match_method`
`PROTECTED_BY`	`match_method`
`SAME_STUDY_AS`	`match_method`
`SPONSORED_BY`	`match_method`
`STUDIES`	`match_method`
`SUBJECT_OF`	`match_method`
`SUBTYPE_OF`	`match_method`
`TARGETS`	`match_method`
`TESTED_IN`	`match_method`
`VARIANT_IN`	`match_method`

Five connect two nodes of the *same* label: `IS_SALT_OF` (122,125), `SAME_STUDY_AS` (32,545), `SUBTYPE_OF` (8,086), `IN_CLASS` 's ATC tree (6,982) and `BIOSIMILAR_OF` (90). None is a self-loop — there are zero edges where src equals dst.

6. Execution order

Order is not cosmetic. Each stage fills dictionaries the next reads; running them out of order produces a graph that validates but resolves against half-built lookups.

Stage	Entry point	Why here
1. Vocabularies	`load_atc`	WHO ATC tree. Loaded whole even for a slice - 1,318 rows, and every substance may point into it.
2. Substance spine	`load_gsrs`	The naming authority. Builds the resolver every later loader matches names through, so nothing needing a name can run before it.
3. Resolver finalised	`r.finalise()`	The stereo tier is built only once every name is known, so a key two different substances both reduce to can be blocked rather than merged.
4. ChEMBL molecules	`load_chembl_molecules`	2.9M molecules, plus the molregno -> node-key map everything ChEMBL needs afterwards.
5. ChEMBL synonyms	`load_chembl_synonyms`	Brand names and research codes, as an alias tier consulted after the identifier tiers.
6. Structures	`load_structures`	InChIKey - chemistry rather than a name, so the strongest merge signal.
7. Reference tables	`reference.ALL`	ATC level 5, salt/parent hierarchy, RXCUI, SPL setids.
8. Targets	`load_targets`	Keyed by UniProt accession where one exists; the tid -> key map is held for mechanisms.
9. Mechanisms	`load_mechanisms`	One file gives both TARGETS and HAS_MECHANISM as stated fact.
10. Disease	`disease.ALL`	MeSH is the spine; chembl_indications carries the MeSH/EFO crosswalk later loaders fold MONDO ids onto. Must precede anything matching disease prose.
11. Products	`products.ALL`	Ten agencies. Also creates all 189 countries and the 9 regions.
12. Trials	`trials.ALL`	Nine registries. WHO last, so native records win on properties.
13. Safety	`safety.ALL`	Regulatory events, FAERS aggregation, VigiAccess organ classes.
14. Variants	`variants.ALL`	Needs symbol_target from HGNC and efo_mesh from chembl_indications.
15. Literature	`literature.ALL`	Needs a finalised resolver and mesh_by_name to match titles against.

7. Every source file, and what it builds

Generated from `graph/sources.py`. 77 files. A trailing `/` means a prefix whose contents are discovered from S3 at build time, so a new file is picked up without a code change.

Clinical Trials Pipeline Intelligence

File	Rows	Builds	Note
`anzctr.org.au/anzctr_trials/anzctr_trials.csv`	40,957	`node:ClinicalTrial`	
`chictr.org.cn/chictr_trails2/chictr_detailed.csv`	217,497	`node:ClinicalTrial`	
`clinicaltrials.gov/clinicaltrials_data/clinicaltrials_all.csv`	606,658	`node:ClinicalTrial`, `node:Company`, `edge:SPONSORED_BY`, `edge:STUDIES`, `edge:TESTED_IN`, `edge:CONDUCTED_IN`	2.7 GB - stream it. conditions and interventions are prose and go through the dictionary matcher, not a direct join.
`clinicaltrialsregister.eu/eu_ctr_trials/eu_ctr_all_trials.csv`	97,822	`node:ClinicalTrial`	
`ctri.nic.in/ctri_trials/ctri_trials.csv`	8,197	`node:ClinicalTrial`	
`euclinicaltrials.eu/ctis_data/CTIS_trials_20260622.csv`	10,158	`node:ClinicalTrial`	
`isrctn.com/isrctn_trials/ISRCTN_search_results.csv`	31,373	`node:ClinicalTrial`	
`jrct.mhlw.go.jp/jrct_trials/jrct_list.csv`	476	`node:ClinicalTrial`	
`trialsearch.who.int/who_trials_csv/who_trials.csv`	1,011,870	`node:ClinicalTrial`, `edge:SAME_STUDY_AS`	4.6 GB. Carries cross-registry ids, so it drives de-duplication. Without it the same study counts once per registry.

Drug Substance Reference

File	Rows	Builds	Note
`atcddd.fhi.no/atc_ddd_data/atc_classes.csv`	1,318	`node:DrugClass`	WHO ATC tree. parent_code gives the hierarchy directly.
`atcddd.fhi.no/atc_ddd_data/atc_substances.csv`	5,678	`edge:IN_CLASS` , `id:ATC`	substance name -> atc_code. Dictionary-matched to Substance.
`dailymed.nlm.nih.gov/dailymed_data/dailymed_master_mapping.csv`	319,885	`id:SPL_SETID` , `id:NDC`	
`ebi.ac.uk-chembl/chembl_data/chembl_action_types.csv`	35	`node:Mechanism`	agonist / antagonist / inhibitor vocabulary.
`ebi.ac.uk-chembl/chembl_data/chembl_indications.csv`	60,504	`edge:INDICATED_FOR`	Carries mesh_id AND efo_id on the same row - the crosswalk that stops opentargets' EFO diseases and MeSH becoming two disconnected populations.
`ebi.ac.uk-chembl/chembl_data/chembl_mechanisms.csv`	7,561	`node:Mechanism` , `edge:TARGETS` , `edge:HAS_MECHANISM`	Carries molregno -> tid AND mechanism_of_action + action_type, so both edges come from one structured file. No inference.
`ebi.ac.uk-chembl/chembl_data/chembl_molecule_hierarchy.csv`	3,508,134	`resolver`	molregno -> parent_molregno. Salt forms as fact: atorvastatin calcium links to atorvastatin without guessing which trailing tokens are salts.
`ebi.ac.uk-chembl/chembl_data/chembl_molecules.csv`	2,877,005	`node:Substance` , `id:CHEMBL_ID` , `node:Modality`	Pan-therapeutic molecule table. molecule_type feeds Modality.
`ebi.ac.uk-chembl/chembl_data/chembl_structures.csv`	3,096,691	`id:INCHIKEY`	InChIKey is the strongest merge signal there is - it is the chemistry, not a name.
`ebi.ac.uk-chembl/chembl_data/chembl_synonyms.csv`	139,044	`resolver`	Extends the name lookup beyond gsrs, notably brand names.
`ebi.ac.uk-chembl/chembl_data/chembl_targets.csv`	18,552	`node:Target`	The Target base. uniprot.org in this lake is disease-scoped (2,782 proteins in themed files), so it cannot be the spine.
`ebi.ac.uk-chembl/chembl_data/chembl_uniprot_mapping.csv`	17,257	`id:UNIPROT`	chembl tid -> UniProt accession, which is the Target key.
`gsrs.ncats.nih.gov/gsrs_data/gsrs_substances.csv`	173,356	`node:Substance` , `id:UNII` , `id:CAS` , `resolver`	The naming authority. preferred_name + synonyms build the name->UNII lookup every other loader resolves through.
`rxnav.nlm.nih.gov/rxnorm_data/rxnorm_drugs.csv`	28,920	`id:RXCUI`	RXCUI is a merge signal and the join to dailymed.

Literature Evidence

File	Rows	Builds	Note
`biorxiv.org/biorxiv/biorxiv_metadata.csv`	2,576	`node:Publication`	Preprints: keyed by DOI, no PMID and no journal.
`europepmc.org/europe_pmc/europe_pmc_metadata.csv`	—	`node:Publication`, `id:PMID`, `id:DOI`, `edge:ABOUT`, `edge:MENTIONS`	66 MB and the largest of the four remaining literature sources. Same 23 columns as pubmed_metadata.csv.
`medrxiv.org/medrxiv/medrxiv_metadata.csv`	1,741	`node:Publication`	As biorxiv.
`pubmed.ncbi.nlm.nih.gov/pubmed/pubmed_metadata.csv`	1,032	`node:Publication`	Same shape as Europe PMC.

MENA GCC Regulatory Market

File	Rows	Builds	Note
`sfda.gov.sa/List_of_Registered_HumanHerbalVeterinary_Drugs.csv`	9,170	`node:Product`, `edge:CONTAINS`, `edge:IN_CLASS`	36 columns incl. scientificName, atcCode1/2, registerNumber. Headers carry a UTF-8 BOM - strip it.
`sfda.gov.sa/Safety_Alert.csv`	260	`node:RegulatoryEvent`	type=safety_alert
`sfda.gov.sa/Shortage_Drugs_List.csv`	1,840	`node:RegulatoryEvent`	type=shortage

Ontologies Standards

File	Rows	Builds	Note
`icd.who.int/icd_data/icd11_codes.csv`	—	`node:Disease`, `id:ICD11`	Second coding system on the same Disease nodes.
`meshb.nlm.nih.gov/mesh_data/mesh_descriptors.csv`	34,154	`node:Disease`, `edge:SUBTYPE_OF`	tree_numbers give the hierarchy: C04.588.322's parent is the descriptor owning C04.588.

Regulatory Approvals

File	Rows	Builds	Note
`accessdata.fda.gov-orangebook/orangebook_data/exclusivity_enriched.csv`	2,265	`node:Exclusivity` , `edge:HAS_EXCLUSIVITY`	
`accessdata.fda.gov-orangebook/orangebook_data/orange_book_unified.csv`	47,497	`node:Product` , `id:FDA_APPL_NO` , `edge:CONTAINS` , `node:Company` , `node:Approval`	Chosen over products.csv: same rows, 18 columns instead of 14, including Applicant_Full_Name and TE_Code.
`accessdata.fda.gov-orangebook/orangebook_data/patents_enriched.csv`	16,344	`node:Patent` , `edge:PROTECTED_BY`	Joins to Product on Appl_No + Product_No. Patent_Use_Code distinguishes a substance patent from a formulation one.
`ema.europa.eu/ema_data/additional_tables/dhpc.csv`	171	`node:RegulatoryEvent` , `edge:SUBJECT_OF`	type=dhpc
`ema.europa.eu/ema_data/additional_tables/orphan_designations.csv`	3,279	`node:RegulatoryEvent` , `edge:SUBJECT_OF`	type=orphan_designation
`ema.europa.eu/ema_data/additional_tables/paediatric_investigation_plans.csv`	3,373	`node:RegulatoryEvent` , `edge:SUBJECT_OF`	type=pip
`ema.europa.eu/ema_data/additional_tables/referrals.csv`	591	`node:RegulatoryEvent` , `edge:SUBJECT_OF`	type=referral
`ema.europa.eu/ema_data/additional_tables/shortages.csv`	82	`node:RegulatoryEvent` , `edge:SUBJECT_OF`	type=shortage
`ema.europa.eu/ema_data/ema_medicines.csv`	2,666	`node:Product` , `id:EMA_PRODUCT` , `edge:CONTAINS` , `edge:IN_CLASS` , `node:Company`	39 columns: active_substance, MAH, atc_code, therapeutic area. Small but dense - centrally authorised EU products.
`health-products.canada.ca/canada_dpd_data/comp.csv`	59,384	`node:Company` , `edge:DEVELOPS`	
`health-products.canada.ca/canada_dpd_data/drug.csv`	59,339	`node:Product`	Canada DPD is already a relational product schema - drug, ingred, comp, route, ther as separate tables. Cleanest source.
`health-products.canada.ca/canada_dpd_data/ingred.csv`	125,922	`edge:CONTAINS`	product -> active ingredients.
`health-products.canada.ca/canada_dpd_data/route.csv`	68,207	`node:Route` , `edge:HAS_ROUTE`	Route vocabulary and the product->route edge in one file.
`health-products.canada.ca/canada_dpd_data/status.csv`	200,119	`node:Approval`	
`health-products.canada.ca/canada_dpd_data/ther.csv`	58,174	`edge:IN_CLASS`	TC_ATC -> DrugClass.
`open.fda.gov/openfda_data/openfda_drugs.csv`	42,173	`node:Product` , `id:NDC`	
`pmda.go.jp/pmda_data/pmda_metadata.csv`	547	`node:Product`	Japan.
`products.mhra.gov.uk/mhra_data/raw_metadata.csv`	78,215	`node:Product` , `id:MHRA_PL`	The UK product population - the same documents the vector store indexes, joined here by licence number.
`purplebooksearch.fda.gov/purplebook_data/patent_list.csv`	424	`node:Patent` , `edge:PROTECTED_BY`	Biologics patents, joined on BLA number.
`purplebooksearch.fda.gov/purplebook_data/purplebook_enriched_products.csv`	2,205	`node:Product` , `node:Approval` , `edge:BIOSIMILAR_OF`	license_type 351(a)/351(k) plus resolved_reference_bla gives the biosimilar-

>originator edge already resolved.

Safety Pharmacovigilance

File	Rows	Builds	Note
`open.fda.gov/Adverse_Events/faers_2020.csv`	—	`node:AdverseEvent` , `edge:HAS_ADVERSE_EVENT`	AGGREGATE at load: one edge per (drug_substance, reaction) with report/serious/death counts. Never one node per report.
`open.fda.gov/Adverse_Events/faers_2021.csv`	—	`node:AdverseEvent` , `edge:HAS_ADVERSE_EVENT`	AGGREGATE at load: one edge per (drug_substance, reaction) with report/serious/death counts. Never one node per report.
`open.fda.gov/Adverse_Events/faers_2022.csv`	—	`node:AdverseEvent` , `edge:HAS_ADVERSE_EVENT`	AGGREGATE at load: one edge per (drug_substance, reaction) with report/serious/death counts. Never one node per report.
`open.fda.gov/Adverse_Events/faers_2023.csv`	—	`node:AdverseEvent` , `edge:HAS_ADVERSE_EVENT`	AGGREGATE at load: one edge per (drug_substance, reaction) with report/serious/death counts. Never one node per report.
`open.fda.gov/Adverse_Events/faers_2024Q1.csv`	—	`node:AdverseEvent` , `edge:HAS_ADVERSE_EVENT`	AGGREGATE at load: one edge per (drug_substance, reaction) with report/serious/death counts. Never one node per report.
`open.fda.gov/Adverse_Events/faers_2024Q2.csv`	—	`node:AdverseEvent` , `edge:HAS_ADVERSE_EVENT`	AGGREGATE at load: one edge per (drug_substance, reaction) with report/serious/death counts. Never one node per report.
`open.fda.gov/Adverse_Events/faers_2024Q3.csv`	—	`node:AdverseEvent` , `edge:HAS_ADVERSE_EVENT`	AGGREGATE at load: one edge per (drug_substance, reaction) with report/serious/death counts. Never one node per report.
`open.fda.gov/Adverse_Events/faers_2024Q4.csv`	—	`node:AdverseEvent` , `edge:HAS_ADVERSE_EVENT`	AGGREGATE at load: one edge per (drug_substance, reaction) with report/serious/death counts. Never one node per report.
`open.fda.gov/Adverse_Events/faers_2025Q1.csv`	—	`node:AdverseEvent` , `edge:HAS_ADVERSE_EVENT`	AGGREGATE at load: one edge per (drug_substance, reaction) with report/serious/death counts. Never one node per report.
`open.fda.gov/Adverse_Events/faers_2025Q2.csv`	—	`node:AdverseEvent` , `edge:HAS_ADVERSE_EVENT`	AGGREGATE at load: one edge per (drug_substance, reaction) with report/serious/death counts. Never one node per report.
`open.fda.gov/Adverse_Events/faers_2025Q3_Q4.csv`	—	`node:AdverseEvent` , `edge:HAS_ADVERSE_EVENT`	AGGREGATE at load: one edge per (drug_substance, reaction) with report/serious/death counts. Never one node per report.
`open.fda.gov/Adverse_Events/faers_2026Q1.csv`	—	`node:AdverseEvent` , `edge:HAS_ADVERSE_EVENT`	AGGREGATE at load: one edge per (drug_substance, reaction) with report/serious/death counts. Never one node per report.
`open.fda.gov/Drug_Recalls/drug_recalls.csv`	17,718	`node:RegulatoryEvent` , `edge:SUBJECT_OF`	type=recall. classification I/II/III is the severity.
`vigiaccess.org/VigiAccess/vigiaccess_adr.csv`	—	`node:OrganClass` , `edge:IN_ORGAN_CLASS`	The MedDRA System Organ Class per reaction - the hierarchy meddra.org itself does not provide. Counts are VigiBase, not FAERS, and are deliberately not merged with FAERS counts.

Targets Genomics Biomarkers

File	Rows	Builds	Note
`cancer.sanger.ac.uk/Cancer_Site_Mutations/`	—	`node:Variant` , `edge:VARIANT_IN`	COSMIC somatic mutations, one file per cancer site. Kept only where the gene is a known drug target.
`genenames.org/data/complete_set/hgnc_complete_set.csv`	42,397	`node:Target`	Enrichment: symbol, name, gene family. uniprot_ids joins to the Target key.
`ncbi.nlm.nih.gov/Variants/variant_summary.csv`	—	`node:Variant` , `id:CLINVAR` , `edge:VARIANT_IN` , `edge:IMPLICATED_IN`	~21.8M rows and NO HEADER ROW - the first line is data, so columns are read by position. Filtered hard: the gene must be a known drug target and the significance must be an actual call.
`platform.opentargets.org/Disease_Associations/Alzheimer_targets.csv`	—	`edge:ASSOCIATED_WITH`	overall_score becomes an edge property. Disease-scoped, so enrichment on top of chembl rather than the base.
`platform.opentargets.org/Disease_Associations/Cancer_targets.csv`	—	`edge:ASSOCIATED_WITH`	overall_score becomes an edge property. Disease-scoped, so enrichment on top of chembl rather than the base.
`platform.opentargets.org/Disease_Associations/Cardiovascular_targets.csv`	—	`edge:ASSOCIATED_WITH`	overall_score becomes an edge property. Disease-scoped, so enrichment on top of chembl rather than the base.

`platform.opentargets.org/Disease_Associations/Diabetes_targets.csv`	—	`edge:ASSOCIATED_WITH`	overall_score becomes an edge property. Disease-scoped, so enrichment on top of chembl rather than the base.
`platform.opentargets.org/Disease_Associations/Infectious_Disease_targets.csv`	—	`edge:ASSOCIATED_WITH`	overall_score becomes an edge property. Disease-scoped, so enrichment on top of chembl rather than the base.
`platform.opentargets.org/Disease_Associations/Respiratory_targets.csv`	—	`edge:ASSOCIATED_WITH`	overall_score becomes an edge property. Disease-scoped, so enrichment on top of chembl rather than the base.
`platform.opentargets.org/Drugs/known_drugs.csv`	203,100	`edge:INDICATED_FOR` , `edge:TARGETS`	Second source for both; chembl is the base.
`uniprot.org/`	—	`id:UNIPROT_ENTRY`	Enrichment only, on targets ChEMBL already established. Disease-scoped, so it cannot be the source of Target nodes - see the EXCLUDED note that used to cover this.

8. What is deliberately not loaded

Source	Reason
`*/Index/*_documents.csv`	Document indexes - filename, URL, product name. These describe the PDFs the vector store already indexes. The graph joins to those documents by product key, so a second copy of the index adds nothing.
`Drug_Substance_Reference/ebi.ac.uk-chembl/chembl_data/chembl_usan_stems.csv`	834 USAN stems. Loaded briefly as Modality and withdrawn: the annotations are not one concept - '-mab' is 'monoclonal antibodies' (modality), '-kinra' is 'interleukin receptor antagonists' (mechanism), '-prazole' is a pharmacologic class - so any single label misfiles most of them. As Moda
`Drug_Substance_Reference/pubchem.ncbi.nlm.nih.gov`	~291M rows across 5 files, but the two big ones are 2-column lookups (CID->SMILES 133.9M, CID->parent 119.1M). They are 85% of the lake's rows and would dominate the graph. chembl_structures already supplies InChIKey, which is the identifier that matters. Revisit only if a PubChem CID crosswalk is specifically needed.
`Literature_Evidence/openalex.org`	openalex 7.9M works, europepmc, pubmed, biorxiv, medrxiv. These build the Publication node, which is Phase 2. Deferred deliberately, not excluded - the openalex file alone is 17.5 GB and would dominate a first build.
`Ontologies_Standards/cdisc.org`	198 rows combined. Standards lists and procurement catalogues with no node to attach to.
`Ontologies_Standards/evs.nci.nih.gov`	NCI Thesaurus, 413k rows. A third disease vocabulary after MeSH and ICD-11. Two coding systems already crosswalk via chembl_indications; a third adds mapping burden without new coverage.
`Ontologies_Standards/loinc.org`	4.4M rows of laboratory observation codes across 56 files. Nothing in the schema consumes them; LOINC describes measurements, not drugs.
`Regulatory_Approvals/ema.europa.eu/.../herbal_medicines.csv`	799 rows. Herbal monographs and veterinary residue limits sit outside the Substance/Product model.
`Regulatory_Approvals/health-products.canada.ca/.../vet.csv`	6,982 veterinary products. Out of scope for a human-medicines graph.
`Regulatory_Approvals/purplebooksearch.fda.gov/.../purplebook_raw_monthly.csv`	The header row is a report title ('Purple Book Monthly Historical Data Changes Report - June 2026'), so the file has no usable column names. purplebook_enriched_products.csv carries the same content parsed.
`Safety_Pharmacovigilance/adrreports.eu`	meddra.org holds no terminology. The scrape reached only public pages - meddra_timeline.csv is news announcements and meddra_versions.csv is release history - because MedDRA itself is licensed. It was planned as the source of the reaction hierarchy and cannot be; vigiaccess publishes reactions already grouped by System Organ Class and is
`Targets_Genomics_Biomarkers/cancer.sanger.ac.uk`	COSMIC gene lists, 40 files / 20k rows. Gene-level cancer census - belongs with a Variant/Gene extension, not the current schema.
`Targets_Genomics_Biomarkers/cbioportal.org`	Planned as a third variant source and is not one. The 23 files are study and clinical metadata - per-patient TCGA cohorts - plus cancer_types.csv, a cancer-type tree keyed by cbioportal's own ids with a `parent` column that references those ids and nothing else. Loading it would add a fourth disease vocabulary that crosswalks to neit
`Targets_Genomics_Biomarkers/ncbi.nlm.nih.gov`	ClinVar variants, ~21.8M rows. There is no Variant node in the schema. Adding one is a real extension (variant -> gene -> target -> disease), not a load - so it is a schema decision first.

9. How the graph meets the vector store

Two stores on two hosts, joined by identifiers rather than by shared storage.

```
Qdrant chunk payload              Graph
  source   ema.europa.eu     <->  Product     agency=EMA
  s3_key   ...PL34771...pdf  <->  Identifier  scheme=MHRA_PL
  doc_type spc | pil | par
  molecule_id  (reserved)    <->  Substance key
```

MHRA document filenames embed the licence number (`ABACAVIRLAMIVUDINE_PL347710263_par.pdf`) and the graph holds 39,002 `MHRA_PL` identifiers — that is the join covering the largest slice of the corpus. The chunk payload reserves `molecule_id` for a direct link, currently unpopulated.

> **SPL setids do not join to the vector store.** DailyMed publishes only CSVs to S3, no documents, so its 393,581 setids identify labels that were never indexed. They remain the canonical handle for a US label and dereference to a DailyMed URL.

Entity lookup uses a second, smaller embedding: 40,432 Disease, DrugClass, AdverseEvent and Mechanism nodes under **SapBERT** (768-d), kept apart from the documents' bge-m3 (1024-d). Full-text handles exact and near-exact names; it cannot bridge *NSCLC* to *Carcinoma, Non-Small-Cell Lung*, which share no characters.